

CS-781

Final Project

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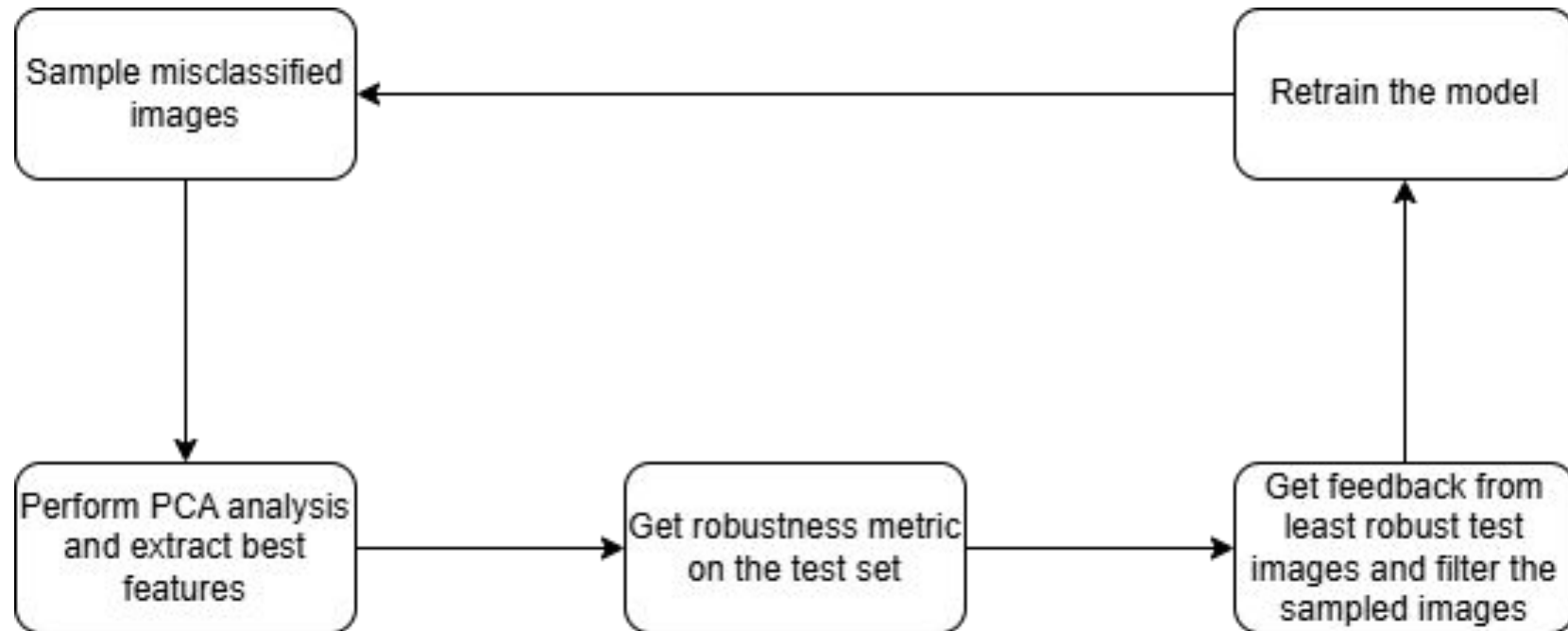
Outline

1. Objective
2. Overall Pipeline
3. Architecture
4. Steps in the Pipeline
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Objective

- The main goal of this project is to employ the VerifAI toolkit to iteratively re-train a convolutional neural network (CNN) to detect if a car is present in the image or not.
- After each re-training iteration, the local robustness of the network is evaluated using the $\alpha\beta$ -CROWN framework.
- As re-training progresses with feedback from local robustness, the CNN is expected to improve in both accuracy and robustness.

Overall Pipeline



Architecture

Layer	Output Shape
Input	3x32x32
Conv1+MaxPool1+ReLU	8x16x16
Conv2+MaxPool2+ReLU	16x8x8
Conv3+MaxPool3+ReLU	32x4x4
Flatten	512
FC1+ReLU	32
FC2	2

- The total number of trainable parameters are **22514**.

Steps in the Pipeline

1. Data Preprocessing and Pre-Training
2. Falsifier and Classifier
3. PCA Analysis
4. $\alpha\beta$ -CROWN Analysis with AutoLiRPA
5. Re-Training

Data Preprocessing and Pre-Training (1/5)

- We first sample 600 images for each class out of which 550 are used for training/validation and the rest of the 50 are used for testing. These will be our training/validation and testing dumps.
- We then pre-train the model as mentioned before with 200 randomly sampled images from the training/validation images for 20 epochs.
- We reshape the original image of size 1242x375 to 32x32 using LANCZOS resampling for training/validation and testing.

Falsifier and Classifier (2/5)

- We begin with the pre-trained model or the model from the previous iteration.
- Using this model, misclassified images are identified with the help of VerifAI.
- We run the falsifier for 600 iterations and sample at most 200 falsified images randomly for each class.

PCA Analysis (3/5)

- For each class PCA analysis is performed on the sampled images and the first principal component is returned.
- Only the continuous space features are considered for this analysis.
- The 2 high-level features with the greatest loadings are saved.

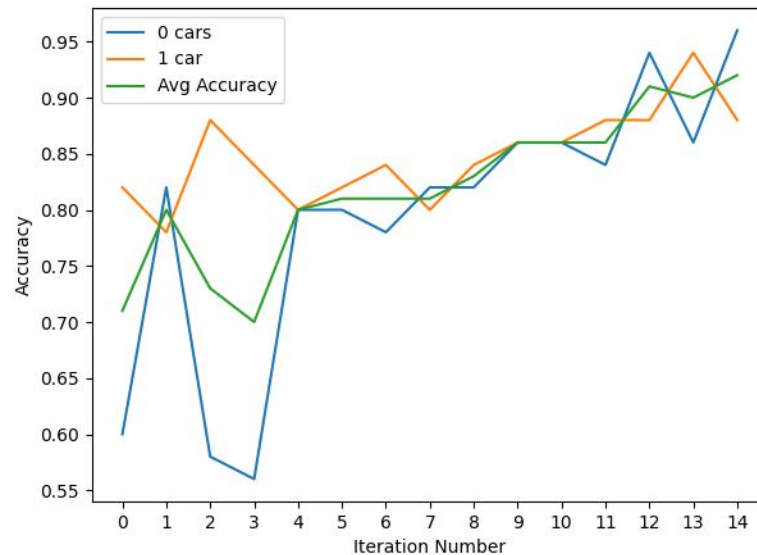
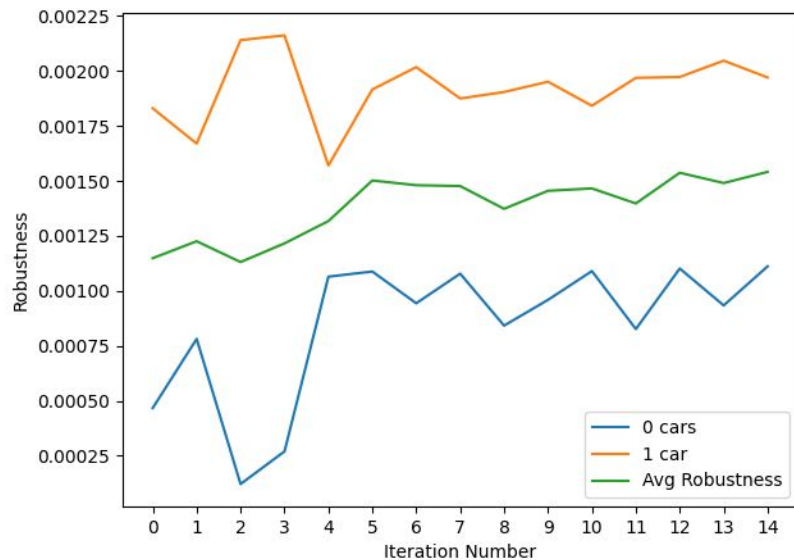
$\alpha\beta$ -CROWN Analysis with AutoLiRPA (4/5)

- We perform $\alpha\beta$ -CROWN analysis on each test image in a class to get their robustness metric values.
- We calculate the largest radius of perturbation around an input image for which the classification output remains unchanged. If the predicted label is not the same as the gold label, negative of the maximum perturbation radius is returned.
- The bottom 20% of test images with the least robustness metric values are taken.
- Then for each high-level feature saved before, the maximum and minimum values are obtained.
- Finally, the sampled images created before are filtered out if those features don't lie in between maximum and minimum values.

Re-Training (5/5)

- For each class, more images are sampled from the original training/validation in the following way:
 - If the number of images sampled from VerifAI is greater than 100, we clip it off to 100 images and randomly sample 100 more images from the original training/validation dump
 - Otherwise we sample enough images from the original training/validation dump to get a total of 200 images.
- Now the previous iteration model or the pre-trained model is re-trained for 5 epochs and saved.

Results



As seen from the figure, robustness and accuracy both increase with iterations.

Thank You